

Tyler Binning

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Skills

Programming: Python, R, SQL, VBA

ML & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Transformers, PEFT

Statistics & Time Series: Linear/Logistic Regression, ANOVA, ARIMA/SARIMA

Visualization: Power BI, Tableau, Matplotlib, Plotly, Streamlit, Dash

Data Engineering: PySpark, Databricks, Docker, MySQL, GCP

Professional Experience

Healthcare Data Analyst

Dec. 2025 - Present

Decatur County Hospital

Leon, IA

- **Analyzed and validated healthcare data** from **EHR/EMR** (Oracle Health) and claims systems to identify trends in patient outcomes, costs, and resource utilization.
- **Developed analytical reports and dashboards (Excel, DA2)** that translated complex clinical and operational data into clear, actionable insights for decision-making.
- **Applied healthcare data standards** (ICD-10, CPT, HL7, FHIR) while ensuring data integrity and compliance with healthcare regulations and reporting requirements.
- **Automated reporting and analysis workflows** within the Microsoft ecosystem using **VBA**, improving efficiency and reducing repetitive manual tasks.

Data Science Intern

Jul. 2025 - Oct. 2025

Planck AI

Remote

- **Led full-cycle AI development**, from data extraction and cleaning to transformer fine-tuning (Phi-2, T5) with LoRA and RAG, boosting accuracy and contextual reasoning.
 - **Built scalable ETL and retrieval systems**, converting legal contract data into JSONL and integrating custom vector stores and prompt templates for efficient document search.
 - **Deployed models in an interactive Streamlit app**, enabling real-time document classification, clause tagging, and analytics visualization.
 - **Advanced model performance through research and collaboration**, applying TF-IDF and Logistic Regression benchmarks, optimizing hyperparameters, and standardizing labeling and evaluation protocols.
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Experiential Learning Projects

Artificial Intelligence Engineering Consultant

Jan. 2025 - Apr. 2025

Statistics & Data Science Consulting | Brigham Young University – Idaho

Rexburg, ID

- **Collaborated** within a team to **design and develop** an AI-powered Teaching Assistant, integrating **generative AI** to provide course guidance and answer student questions.
 - **Integrated** course materials into the AI Teaching Assistant using a Retrieval-Augmented Generation (**RAG**) model, ensuring accurate, context-aware responses to enhance student learning.
 - **Worked with faculty** to refine **AI-generated insights**, ensuring alignment with academic standards and improving instructional effectiveness.
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Education

Bachelor of Science in Data Science, *Brigham Young University - Idaho*

Sept. 2022 - Dec 2025

- Data Science Society – Vice President